Chapter 7

Lesson 28-11-2025

7.1 Orthogonal Frequency Division Multiplexing (OFDM)

7.1.1 OFDM Transmitter

Notation Change and "Blockization"

In OFDM, we change the standard notation compared to single-carrier systems. Usually, input symbols are indicated as $x[n]$ in the time domain. In OFDM, we denote the input symbols as $x[k]$. The k represents the index in the **frequency domain**, since these symbols will be transmitted on different subcarriers.

The input symbols are grouped ("packed") into blocks of size K.

$$\mathbf{x} = [x[0], x[1], \dots, x[K-1]] \tag{7.1}$$

These symbols may belong to a complex constellation, for example, 256-QAM.

> **❓ Student Q&A: Block Stretching**
> **Q:** What is meant by "stretching" the blocks?
> **A:** During the lecture, a student asked for clarification on "lengthening" the blocks. The professor explained that to transmit a block of K symbols, if one wants to maintain the same input rate, time or bandwidth must be increased. However, in the OFDM perspective, the key concept is that we take K symbols and transmit them *in parallel* over a longer time period compared to a single symbol, occupying a bandwidth $B \approx K/T$ (where T is the block time). Technically, in the context of the explanation, "stretch" referred to the insertion of the Cyclic Prefix, which physically lengthens the duration of the transmitted block from K to $K+L$ samples.

IDFT (Inverse Discrete Fourier Transform)

The fundamental operation at the transmitter is the IDFT applied to each block of K symbols. This step brings the symbols from the frequency domain to the time domain. The formula to generate the signal in the time domain $x[n]$ is:

$$x[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j\frac{2\pi}{K}nk} \quad \text{for } n = 0, \dots, K-1 \tag{7.2}$$

Cyclic Prefix (CP) Insertion

To mitigate multipath channel effects, a Cyclic Prefix is inserted. If the length of the channel impulse response is L, we insert a prefix of length L (or greater) at the beginning of the block. The extended vector $\overline{x}$ will have dimension $K + L$. The operation consists of copying the last L samples of the block $x[n]$ and placing them at the beginning:

$$\overline{x} = [\underbrace{x[K-L], \dots, x[K-1]}_{\text{Cyclic Prefix (copy)}}, \underbrace{x[0], x[1], \dots, x[K-1]}_{\text{Original Block}}] \tag{7.3}$$

Formally, the negative indices in the prefix correspond to the final indices of the original block: $x[-1] \leftarrow x[K-1]$, $x[-2] \leftarrow x[K-2]$, and so on.

Cyclic Property: The insertion of the CP makes the signal "circular" with respect to the block length K. Mathematically, thanks to the periodicity of the complex exponential, the following relationship holds:

$$x[-1] = x[K-1] \tag{7.4}$$

In fact, $e^{j\frac{2\pi}{K}(-1)k} = e^{j\frac{2\pi}{K}(K-1)k}$.

7.2 The Channel

The transmitted signal passes through a channel modeled as a Finite Impulse Response (FIR) filter with coefficients $h[l]$ of length $L + 1$ ($l = 0 \dots L$). The channel output $y[n]$ is given by the linear convolution between the transmitted signal (including the CP) and the channel response:

$$y[n] = \sqrt{G} \sum_{l=0}^{L} h[l]\overline{x}[n-l] \tag{7.5}$$

Inter-Block Interference (IBI): For indices corresponding to the cyclic prefix (e.g., $n = -1, \dots, -L$), the convolution "picks up" symbols from the previous transmitted block. For example, for $y[-1]$, the summation involves $\overline{x}[-1-L]$, which belongs to the previous block.

❓ Student Q&A: CP Constraints
Q: Is there a constraint on the length of the Cyclic Prefix?
A: Yes. The length of the Cyclic Prefix (L) must be greater than or equal to the length of the channel impulse response. This ensures that all Inter-Block Interference (IBI) falls within the cyclic prefix, which is then discarded at the receiver, leaving the data block "clean".

7.3 The OFDM Receiver

The receiver must recover the original symbols $x[k]$. The operations are specular to those of the transmitter.

7.3.1 Cyclic Prefix Removal

The first L received samples ($y[-L], \dots, y[-1]$) are contaminated by interference from the previous block. The solution is simple: discard the cyclic prefix. We consider only the vector $\boldsymbol{y}$ of length K:

$$y = [y[0], y[1], \dots, y[K-1]] \tag{7.6}$$

7.3.2 DFT (Discrete Fourier Transform)

On the vector cleaned of the cyclic prefix, we apply the DFT:

$$\tilde{y}[n] = \sum_{p=0}^{K-1} y[p] e^{-j\frac{2\pi}{K}np} \tag{7.7}$$

Here n represents the subcarrier index at reception.

7.4 Mathematical Proof [The ”Heart” of OFDM]

Let us see why OFDM works by substituting the channel expression into the DFT formula.

7.4.1 Step 1: Expansion of $\tilde{y}[n]$

Substitute $y[p]$ with the channel convolution:

$$\tilde{y}[n] = \sum_{p=0}^{K-1} \left(\sqrt{G} \sum_{l=0}^{L} h[l] \overline{x}[p-l] \right) e^{-j\frac{2\pi}{K}np} \tag{7.8}$$

7.4.2 Step 2: Substitution of x (IDFT)

Substitute $\overline{x}[p-l]$ with its IDFT definition (the original input $x[k]$):

$$\tilde{y}[n] = \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^{L} h[l] \left(\frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j\frac{2\pi}{K}(p-l)k} \right) e^{-j\frac{2\pi}{K}np} \tag{7.9}$$

Note: Thanks to the cyclic prefix, the time shift $p-l$ in the linear convolution acts as a circular shift, allowing the use of the IDFT complex exponentials.

7.4.3 Step 3: Rearrangement of Summations

We bring out the constants and group the terms:

$$\tilde{y}[n] = \frac{\sqrt{G}}{K} \sum_{k=0}^{K-1} x[k] \left(\sum_{l=0}^{L} h[l] e^{-j\frac{2\pi}{K}lk} \right) \left(\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)} \right) \tag{7.10}$$

Let us analyze the two terms in parentheses:

1. **Channel DFT ($\tilde{h}[k]$):** The term $\sum_{l=0}^{L} h[l]e^{-j\frac{2\pi}{K}lk}$ is exactly the frequency response of the channel at frequency k, which we call $\tilde{h}[k]$. We assume $h[l] = 0$ for $l > L$.
2. **Orthogonality:** The term $\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)}$ sums complex exponentials. This sum is non-zero only if $k = n$.

$$\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)} = K \cdot \delta(k-n) \tag{7.11}$$

where $\delta(x) = 1$ if $x = 0$, otherwise 0.

7.4.4 Final Result

Thanks to the Dirac delta $\delta(k-n)$, the summation over k "survives" only for the term where $k = n$. The factor K in the numerator cancels with $1/K$. We obtain:

$$\tilde{y}[n] = \sqrt{G} \cdot \tilde{h}[n] \cdot x[n] \tag{7.12}$$

> **❓ Student Q&A: Notation Confusion**
> **Q:** Confusion regarding n vs k notation in the final proof.
> **A:** Some ambiguity was noted in the use of indices. In the final formula, n is used as the output subcarrier index ($\tilde{y}(n)$). The formula proves that the output at index n depends *only* on the input at the same index n ($x[n]$), scaled by the channel $h[n]$. This confirms the absence of Inter-Carrier Interference (ICI).

7.4.5 Interpretation and Equalization

The result $\tilde{y}[n] = \sqrt{G}\tilde{h}[n]x[n]$ is fundamental. It means that, in the frequency domain, there is no interference between symbols or subcarriers. The symbol transmitted on frequency n ($x[n]$) is merely scaled by a complex coefficient $\tilde{h}[n]$ (the channel response at that frequency).

One-Tap Equalization: To recover the transmitted symbol $\hat{x}[n]$, a simple division (Zero-Forcing) is sufficient, assuming the channel is known:

$$\hat{x}[n] = \frac{\tilde{y}[n]}{\sqrt{G} \cdot \tilde{h}[n]} \tag{7.13}$$